
 LODE AND LODI, TWO NEW INITIAL VALUE ORDINARY DIFFERENTIAL EQUATION SOLVERS *

Alan C. Hindmarsh
 Mathematics and Statistics Division
 Lawrence Livermore Laboratory
 Livermore, California 94550

Two new packages are available for the numerical solution of the initial value problem for stiff and nonstiff systems of ordinary differential equations (ODE's). LODE solves explicitly given ODE systems, while LODI solves systems given in linearly implicit form, including differential-algebraic systems. Both are based on older packages that have proved highly popular, but both reflect a wide variety of improvements in the design of the user interface, algorithms, and program structure.

LODE

LODE (Livermore Solver for Ordinary Differential Equations) is based on the GEAR and GEARB packages. It solves ODE systems given explicitly as $dy/dt = f(t,y)$. Some of the important new features of LODE are the following:

1. By use of a method flag, LODE handles either a full or banded Jacobian matrix ($J = \partial f / \partial y$), where before two separate packages were needed for the two cases. (The use of J arises with stiff systems only.)

2. The user interface (call sequence, etc.) has been completely redesigned in an attempt to achieve both maximum ease of use and maximum flexibility. (See the section below on background). While the basic methods in LODE are the same as in GEAR and GEARB, many new capabilities have been added, and many others have been included in such a way that they are much easier to activate than before. A few examples are:

- user names for f and J routines passed as arguments,
- work arrays to allow for flexibility of problem parameters without requiring recompilation,
- internally computed initial step size,
- more flexible error control parameters (4 scalar/vector tolerance choices),
- independent flags for starting and stopping options,
- easy changing of input parameters in mid-problem,

- convenient optional inputs (e.g., maximum method order) and optional outputs (e.g., step and function evaluation counts),
- optional provision of derivatives of the solution (of various orders) at any point,
- real and integer user data space (of dynamic length) available in f and J routines, with no extra burden on the casual user.

3. The user documentation, which is contained in the initial comment cards of the source, is given in a two-level form. A short and simple set of instructions, with a short example program, is given first, for the casual user. Then detailed instructions are given for users with special problem features or a desire for nonstandard options. The latter is also organized so as to allow selective reading by a user who wants only a fraction of the nonstandard capabilities. A separate user's manual is in process.

4. When stiff options are selected, linear systems are solved with routines from LINPACK, which is becoming a widely accepted standard collection of linear system solvers.

5. Some retuning of various heuristics was done so that performance should be more reliable than for GEAR/GEARB.

6. The core routine which takes a single step, called STODE, is independent of the way in which the Jacobian matrix (if used) is treated. Thus as variant versions of LODE are written to use other matrix structures, a single Subroutine STODE will be used for all.

7. The writing of all error messages is done in a small isolated general-purpose message handler called XERRWV. Two other small subroutines are user-callable for optional changing of the output unit number and optional suppression of messages. This trio of routines is compatible with a much larger error package (the SLATEC Error Handling Package) written elsewhere.

8. LODE easily allows a user to interrupt a problem and restart it later (e.g. in switching between two or more ODE problems). Also, using LODE in overlay mode is very easy, with no loss of needed local variables.

9. The various lists of constants needed for the integration, formerly appearing in Subroutine COSET, are now computed (once per problem). This adds to the portability of LODE.

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LSODI

LSODI is a variant of LSODE, written primarily by Jeffrey F. Painter at LLL, for ODE systems given in linearly implicit form $A \, dy/dt = g(t,y)$, where A is a square matrix. LSODI allows A to be singular, but in that case it requires the user to input consistent initial values of both y and dy/dt . LSODI is also based (in part) on the GEARIB package, but corrects a number of basic deficiencies, as follows:

1. The matrices involved can be treated as either full or banded, by use of the method flag.
2. The dependence of A on y is automatically and inexpensively accounted for, whether partial derivatives are supplied by the user or computed internally by difference quotients.
3. When A is singular, the user needs to supply only the initial value of dy/dt , and this array (along with the initial y) is passed through the call sequence, rather than computed in a user-replaceable package routine. (Admittedly, correct initial data can be difficult to obtain for some type of problems.) When the initial dy/dt is not being supplied, an input flag instructs LSODI to compute it on the assumption that A is initially nonsingular. Thereafter, no such assumption is made, but ill-conditioning can be a problem when A is singular.
4. The user-supplied residual routine includes a flag which allows the user to signal either an error condition or an interrupt condition.
5. All of the features described above under LSODE are shared by LSODI, with a few minor variations as dictated by differences in the type of problem being solved.

The modular and systematized organization of both LSODE and LSODI allows most of the routines in each package to be shared between them. Other variants to be written in the future will share this property.

BACKGROUND

The design of LSODE and LSODI was part of a sequence of discussions and efforts aimed at setting standards for ODE solvers, and was sponsored by the Applied Mathematical Sciences Research Program under the Office of Basic Energy Sciences in the U.S. Dept. of Energy. The long-range goal of these efforts is a systematized collection of high-quality ODE initial value solvers, tentatively called ODEPACK. A short-range goal initially was to set standards for the user interface of such solvers. In 1978, workshops and discussions involving various people (mostly at DOE laboratories) in the ODE software field, and also involving users of such software, led to a tentative standard, documented in LLL Report UCID-17954, A Tentative User Interface Standard for ODEPACK, by A. C. Hindmarsh (October 1978). The following people (listed

alphabetically) contributed in varying degrees to this interface design effort: Bill Buzbee, George Byrne, Jim Cody, Kirby Fong, Fred Fritsch, Pat Gaffney, Dick Hickman, Alan Hindmarsh, Bob Huddleston, Tom Jefferson, Ron Jones, Dave Kahaner, Bob Kee, Jim Pool, Mel Scott, Larry Shampine, Dexter Sutherland, and Buddy Watts. In the process, an agreement was also made to rewrite some of the more popular ODE solvers in conformity with this tentative user interface standard. LSODE is the result of that agreement in the case of the GEAR and GEARB packages. LSODI was then generated in the same spirit, although the implicit ODE problem was not within the scope of the original plans. In the meantime, unfortunately, some of the other participants reversed themselves and withdrew from the effort, and as a result there are no corresponding rewrites of other solvers at this time.

AVAILABILITY

Both the LSODE and LSODI packages are available on request, free of charge. Each is available in both a single and a double precision version. A blank magnetic tape should be sent with the request, and with desired tape specifications noted. The standard specifications at LLL are:

7-track even parity BCD
or 9-track odd parity EBCDIC
800 BPI, 80 characters per record
No blocking or header label

but minor variations can be accommodated.

The LSODE package is also in the process of being installed at the National Energy Software Center (formerly the Argonne Code Center), and requests for it are now being accepted by NESC. NESC will then cease supplying GEAR or GEARB. NESC operates on a subscription basis, and requestors should work through their installation representatives.

To submit a direct request or obtain additional information, please contact:

Alan C. Hindmarsh
Mathematics and Statistics Division
L-300
Lawrence Livermore Laboratory
P. O. Box 808
Livermore, CA 94550
(415) 422-4276
FTS: 532-4276